

Hidden "In-Between" Materials May Aid Batteries and Solar Fuels

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When chemists heat ingredients to make a solid material, they usually focus on the finished product. A new study says the brief stages in the middle can matter just as much.

Researchers reported in *Nature Communications* that they watched specially designed starter molecules, called single-source **precursors**, change as temperature rose. By capturing those temporary forms, they found previously unknown material phases that normal methods can miss.

Dr. Sebastian Pike of the University of Warwick said scientists often think only about going from A to B. This work shows many interesting steps between A and B, and some of those hidden steps may be useful on their own.

One discovery was a new, kinetically stabilized form of **bismuth vanadate**, named γ -BiVO₄. Ordinary BiVO₄ is already studied for clean energy because its **band gap** the energy needed to absorb sunlight and drive reactions helps it use light to help split water and make hydrogen fuel. The new γ form has a different atomic arrangement and a larger **band gap**, so it interacts with light differently. That could give engineers new options for solar fuels, catalysts, and electronics.

Another **intermediate** found in the experiments could store large amounts of **lithium**, which raises interest for future batteries. Dr. Dominik Kubicki of the University of Birmingham said these in-between materials are not only stepping stones; they can have useful properties themselves.

The team used tools such as solid-state NMR spectroscopy, X-ray diffraction, and pair distribution function analysis to see structures that usually appear only briefly. They also found that the starting precursor and how it breaks apart strongly affect which materials form. Pike said studying only a few **precursors** already suggests a wider chance to find more hidden but useful materials by carefully controlling heat and chemistry.

Word Treasure

precursors -- Starter chemicals that turn into a final material when heated.

band gap -- The energy a material needs to absorb light and start useful reactions.

bismuth vanadate -- A compound studied for turning sunlight into clean-energy chemistry.

lithium -- A light metal widely used in rechargeable batteries.

intermediate -- A temporary form that appears while a material is still changing.

Background you need

Chemists often heat carefully designed molecules to build solid materials for energy technology.

Solar-fuel research looks for materials that use sunlight to drive useful chemical reactions such as making hydrogen.

Battery materials need to store and release charged particles like lithium efficiently.

Advanced lab tools can catch short-lived structures that ordinary recipes miss.

Break it down

LEAD: Hidden heating stages can yield useful new materials

KEY EVENT: Team watches precursors transform and catches unknown phases

+ - FIND: New γ -BiVO₄ with different light behavior

+ - FIND: Intermediate that stores much lithium

METHODS: NMR, X-ray diffraction, pair distribution analysis

WHY: Better control of pathways may unlock hard-to-make structures

OPEN QUESTION: How many more useful intermediates exist

Why it matters

Clean energy and better batteries depend on discovering materials that work well with light and charged particles. Looking at temporary stages gives scientists more places to find helpful inventions.

Test what you remember

1. Where was the research reported?

- ☐ (A) A sports magazine
- ☐ (B) A weather app
- ☐ (C) A cooking blog
- ☐ (D) Nature Communications

2. What new form of BiVO₄ did they name?

- ☐ (A) Oblik-4
- ☐ (B) Crew-12
- ☐ (C) ?-BiVO₄
- ☐ (D) Alpha foam

3. Why is BiVO₄ studied for clean energy?

- ☐ (A) It cools the Moon
- ☐ (B) Its band gap helps it use sunlight in useful reactions
- ☐ (C) It makes headphones louder
- ☐ (D) It is used as rocket fuel only

4. What metal storage ability was noted for another intermediate?

- ☐ (A) Only wood
- ☐ (B) Only gold
- ☐ (C) Large amounts of lithium
- ☐ (D) Only helium

5. Who is Dr. Sebastian Pike affiliated with in the article?

- ☐ (A) NASA Johnson
- ☐ (B) University of Warwick
- ☐ (C) NPR
- ☐ (D) Virginia Tech

6. What do the scientists say about "in-between" stages?

- ☐ (A) They are always dangerous waste
- ☐ (B) They never matter
- ☐ (C) They only happen on Mars
- ☐ (D) They can have useful properties of their own

Answer key (try the quiz first!): 1: D 2: C 3: B 4: C 5: B 6: D

Questions to think about

1. **Dr. Sebastian Pike:** Says hidden steps between start and finish can be as important as the final product.
2. **Dr. Dominik Kubicki:** Notes intermediate materials can have useful properties in their own right.

MY THOUGHTS (at least 20 words):